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(a) Backbone reconstruction from distance data for PDB 6LU7

I downloaded the PDB structure **6LU7** and extracted the **Cα coordinates of residues 102–152 of chain A**. This gave a set of:

$$n = 152-102+1 = 51$$

Cα atoms. I used the Cα atom of residue 102 as the reference atom and placed it at the origin. Therefore, the Gram matrix was constructed for the remaining 50 atoms, so:

$$G \in \mathbb{R}^{50\times 50}$$

The pairwise Euclidean distances were computed as:

$$d_{ij} = \|x_i - x_j\|$$

and the Gram matrix was constructed from the distance data using:

$$G_{ij} = \frac{d_{0i}^2 + d_{0j}^2 - d_{ij}^2}{2}$$

where atom 0 is the reference Cα atom of residue 102.

The SVD of GGG gave the following leading singular values:

$$\sigma_1 = 1.404837902334 \times 10^4$$

$$\sigma_2 = 3.619412961033 \times 10^3$$

$$\sigma_3 = 9.173617401326 \times 10^2$$

$$\sigma_4 = 1.270445143734 \times 10^{-12}$$

The remaining singular values were also approximately 10^{-12}. Using a tolerance of 10^{-8}, the numerical rank was therefore:

rank(G) = 3

This is expected because the Gram matrix satisfies:

$$G = XX^T$$

where:

$$X \in \mathbb{R}^{50\times 3}$$

contains the coordinates of the 50 non-reference Cα atoms relative to residue 102. Therefore:

$$\mathrm{rank}(G) \leq 3$$

Since the selected protein fragment is genuinely three-dimensional and not degenerate into a line or a plane, the rank is exactly 3.

The coordinates were reconstructed from the top three singular values and singular vectors using:

$$X_{\mathrm{rec}} = U_3 \Sigma_3^{1/2}$$

Then the reference atom was added back at the origin, giving a reconstructed set of 51 Cα coordinates.

The reconstructed structure was aligned to the original structure using an optimal orthogonal alignment. The determinant of the optimal alignment matrix was:

$$\det(R) = -1$$

This indicates that the SVD reconstruction produced the reflected version of the original structure. This is expected because distance data determine the geometry up to translation, rotation, and reflection. Distances are invariant under reflection, so the mirror image has exactly the same pairwise distances as the original structure.

After optimal alignment allowing this reflection, the coordinate RMSD was:

$$c\text{-RMSD} = 8.987443256654 \times 10^{-15}\ \text{Å}$$

This value is essentially zero, showing that the reconstructed structure has the same geometry as the original Cα backbone, up to an orthogonal transformation.

Conclusion: The Gram matrix has rank 3 because the Cα atoms lie in three-dimensional Euclidean space. The SVD reconstruction from distance data correctly recovers the original backbone geometry. The final c-RMSD after alignment is approximately zero, confirming the correctness of the reconstruction.

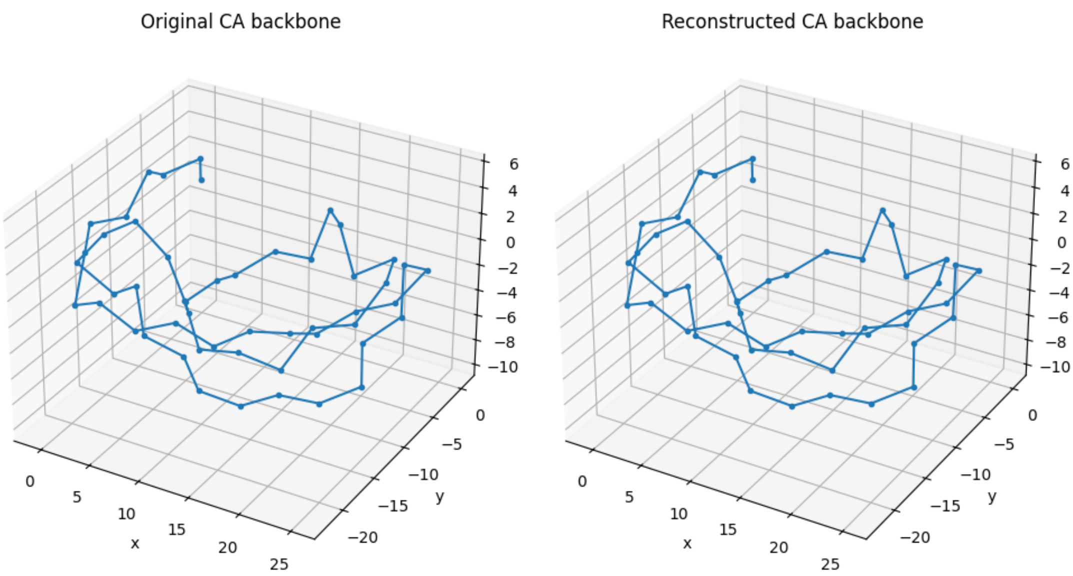

Figure 1. Original and reconstructed Cα backbone for residues 102–152 of chain A in PDB structure 6LU7. Left: original Cα backbone after placing residue 102 at the origin. Right: reconstructed Cα backbone after Gram matrix reconstruction and optimal alignment. The two structures have the same geometry, with final c-RMSD = $8.987443256654\times10^{-15}$ Å.

(b) Reconstruction after adding random noise to pairwise distances

Parameter	Value
PDB structure	6LU7
Chain	A
Atoms used	Cα atoms only
Residues	102–152
Number of Cα atoms	51
Reference atom	Cα of residue 102
Noise level	random relative noise up to ±10%
Random seed	42
Rank tolerance	10^-8

Method

I started from the same Cα coordinate set used in the noise-free case: residues 102–152 of chain A from PDB structure 6LU7. This gives 51 Cα atoms. The Cα atom of residue 102 was used as the reference atom, so the Gram matrix was constructed for the remaining 50 atoms and has size 50 × 50.

First, I computed the exact pairwise Euclidean distances d_ij between all Cα atoms. Then I added independent random relative noise of up to 10% to every off-diagonal distance while preserving the original atom numbering along the backbone:

$$d_{ij}^{\text{noisy}} = d_{ij}(1+\epsilon_{ij})$$

where:

$$\epsilon_{ij} \sim U[-0.1, 0.1]$$

The noisy distance matrix was kept symmetric, meaning d_ij^noisy = d_ji^noisy, and its diagonal was set to zero because the distance of each atom from itself is zero.

Using the noisy distances, I constructed the noisy Gram matrix with the same formula as in the exact case:

$$G_{ij}^{\text{noisy}} = \frac{(d_{0i}^{\text{noisy}})^2 + (d_{0j}^{\text{noisy}})^2 - (d_{ij}^{\text{noisy}})^2}{2}$$

Here, atom 0 is the reference Cα atom of residue 102. I then computed the SVD of this noisy Gram matrix and reconstructed the coordinates using only the three largest singular values:

$$X_{\text{rec}} = U_3 \Sigma_3^{1/2}$$

Finally, I aligned the reconstructed coordinates to the original Cα coordinates and computed the coordinate RMSD (c-RMSD). Reflection was allowed during the optimal alignment, because distance data are invariant under reflection and therefore cannot distinguish a structure from its mirror image.

Results

Quantity	Exact distances	Noisy distances
Gram matrix size	50 × 50	50 × 50
Numerical rank, tolerance 10^-8	3	50
det(R) after alignment	-1.000000000000	-1.000000000000
c-RMSD after alignment	8.987443256654 × 10^-15 Å	1.447742892514 Å

For the noisy Gram matrix, the first ten singular values were:

Singular value	Value
σ1	1.435688166077 × 10^4
σ2	3.582918306419 × 10^3
σ3	9.451727291780 × 10^2
σ4	4.018805718314 × 10^2
σ5	3.635405381449 × 10^2
σ6	2.511821499430 × 10^2
σ7	2.468684339519 × 10^2
σ8	2.152762183889 × 10^2
σ9	2.135717664965 × 10^2
σ10	2.088691420459 × 10^2

Explanation of the rank

In the exact-distance case, the Gram matrix has rank 3 because it can be written as G = XX^T, where X contains the coordinates of the 50 non-reference atoms in R^3. Therefore rank(G) ≤ 3, and since the fragment is genuinely three-dimensional, the rank is exactly 3.

After adding random noise of up to 10% to each pairwise distance, the distances are no longer perfectly consistent with a set of points in three-dimensional Euclidean space. In other words, the noisy Gram matrix is no longer exactly expressible as XX^T with X in R^(50×3). As a result, many singular values beyond the first three become numerically non-zero, and the numerical rank increases to 50 using a tolerance of 10^-8.

This does not mean that the final reconstructed coordinates have 50 dimensions. The reconstruction was explicitly truncated to the three largest singular values, producing a three-dimensional approximation to the noisy distance data.

<u>c-RMSD interpretation</u>

Using the three largest singular values, the reconstructed noisy structure was aligned with the original Cα backbone. The final c-RMSD was 1.447742892514 Å. This value is no longer approximately zero, unlike the exact-distance case, because the input distances were randomly perturbed. Therefore, the reconstructed structure is only an approximation of the original backbone geometry.

The determinant of the optimal alignment matrix was det(R) = -1. This indicates that the optimal alignment includes a reflection. This is expected and acceptable in this distance-based reconstruction, because pairwise distances are invariant under reflection.

<u>Conclusion</u>

Adding random relative noise of up to ±10% to the pairwise distances destroys the exact rank-3 Euclidean structure of the original distance data. Consequently, the noisy Gram matrix becomes numerically full rank, with rank 50 at tolerance 10^-8. By retaining only the three largest singular values, I obtained the best rank-3 reconstruction used for a 3D coordinate approximation. After optimal alignment to the original structure, the c-RMSD was 1.447742892514 Å, showing that the reconstruction remains close to the original backbone but is no longer exact because of the introduced noise.

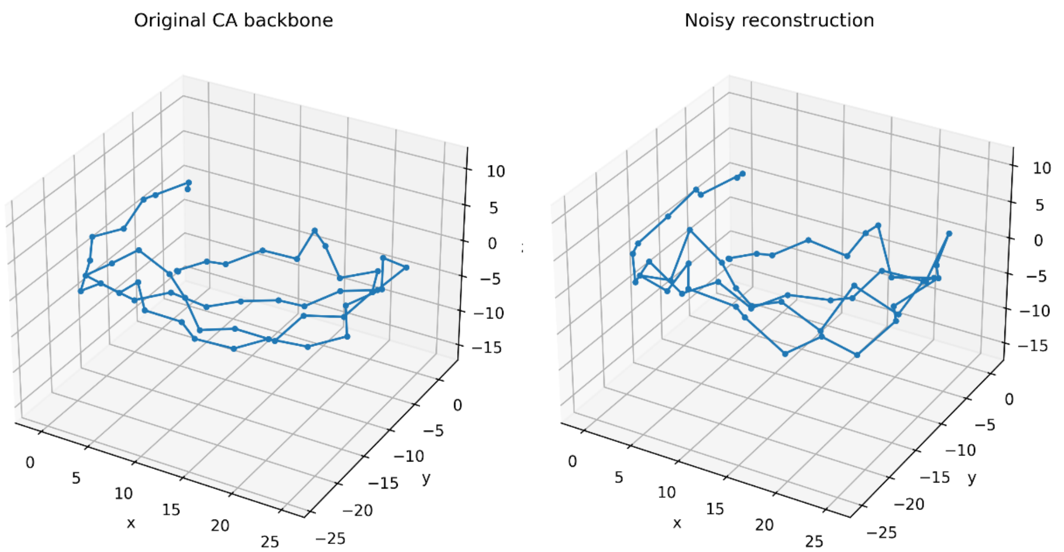

Figure 2. Original and noisy reconstructed Cα backbone for residues 102–152 of chain A in PDB structure 6LU7. Left: original Cα backbone after placing residue 102 at the origin. Right: reconstructed backbone obtained from noisy distances after retaining the three largest singular values and applying optimal alignment. The final c-RMSD was 1.447742892514 Å.

(c) Reconstruction with another reference atom using the same noisy distances

I repeated part (b) using the same noisy distance matrix, but changed the reference atom used to construct the Gram matrix. In part (b), the reference atom was atom 1, corresponding to Python index 0 and residue 102 Cα. In this part, I used atom 25 as the new reference atom. Since Python indexing starts from 0, atom 25 corresponds to Python index 24, i.e. residue 126 Cα in the selected fragment residues 102–152.

The same noisy distances were used in both reconstructions. The noise was generated once using random seed 42, so the comparison isolates the effect of changing only the reference atom and not the noise realization.

$$d_{ij}^{\text{noisy}} = d_{ij}(1 + \varepsilon_{ij}), \quad \varepsilon_{ij} \in [-0.1, 0.1]$$

Construction of the Gram matrix

For the new reference atom, the Gram matrix was constructed using the same distance-geometry formula, but with r denoting the selected reference atom:

$$G_{ij} = \frac{d_{ri}^2 + d_{rj}^2 - d_{ij}^2}{2}$$

For reference atom 25, r is the Cα atom of residue 126. The reference atom was placed at the origin, and the Gram matrix was constructed for the remaining 50 Cα atoms. Therefore, in both reference choices, the Gram matrix had shape 50 × 50.

SVD, rank, and coordinate reconstruction

The SVD of each noisy Gram matrix was computed, and the numerical rank was defined as the number of singular values larger than 10^{-8}. For both choices of reference atom, the noisy Gram matrix was numerically full rank:

$$\text{rank}(G_{\text{ref1}}) = 50$$

$$\text{rank}(G_{\text{ref25}}) = 50$$

This is expected. With exact distances, changing the reference atom only changes the coordinate frame and does not change the underlying geometry. However, with noisy distances, the distance matrix is no longer perfectly embeddable in R^3. Therefore, the corresponding noisy Gram matrix is not exactly rank 3, and the best rank-3 reconstruction may depend slightly on the chosen reference atom.

The reconstructed coordinates were obtained using only the three largest singular values and their corresponding singular vectors:

$$X_{\text{rec}} = U_3 \Sigma_3^{1/2}$$

This gives the best rank-3 approximation used for reconstructing a three-dimensional structure from the noisy Gram matrix.

Numerical results

Case	Reference atom	Reference residue	Rank	c-RMSD after alignment
Noisy distances	Atom 1 / Python index 0	Residue 102 Cα	50	1.447742892514 Å
Noisy distances	Atom 25 / Python index 24	Residue 126 Cα	50	1.397538651934 Å

For reference atom 1, the first ten singular values were:

sigma_1 = 1.435688166077e+04

sigma_2 = 3.582918306419e+03

sigma_3 = 9.451727291780e+02

sigma_4 = 4.018805718314e+02

sigma_5 = 3.635405381449e+02

sigma_6 = 2.511821499430e+02

sigma_7 = 2.468684339519e+02

sigma_8 = 2.152762183889e+02

sigma_9 = 2.135717664965e+02

sigma_10 = 2.088691420459e+02

For reference atom 25, the first ten singular values were:

sigma_1 = 5.162083036483e+03

sigma_2 = 3.413861975140e+03

sigma_3 = 1.569675636705e+03

sigma_4 = 4.369198527072e+02

sigma_5 = 2.919894301286e+02

sigma_6 = 2.595662218684e+02

sigma_7 = 2.316897683895e+02

sigma_8 = 2.233832999434e+02

sigma_9 = 2.232501311356e+02

sigma_10 = 2.024473238458e+02

c-RMSD comparison

After optimal alignment with the original Cα coordinates, the c-RMSD values were:

$$c\text{-RMSD}_{\text{ref1}} = 1.447742892514\ \text{Å}$$

$$c\text{-RMSD}_{\text{ref25}} = 1.397538651934\ \text{Å}$$

$$\Delta c\text{-RMSD} = 0.050204240580\ \text{Å}$$

The c-RMSD obtained with reference atom 25 was slightly lower than the c-RMSD from part (b), where atom 1 was used as the reference. The difference was only approximately 0.0502 Å, so the two reconstructions are very close. They are not expected to be identical, because the noisy distances do not define one exact three-dimensional Euclidean embedding.

Geometric ambiguity of the reconstruction

The reconstruction from distances is geometrically ambiguous in absolute coordinates. Pairwise distances determine the structure only up to translation, rotation, and reflection. Therefore, the reconstructed coordinates do not have to match the original PDB coordinates directly before alignment.

In both reference-atom reconstructions, the determinant of the optimal alignment matrix was:

$$\det(R) = -1$$

This means that the optimal orthogonal alignment included a reflection. This is not an error: distances are invariant under reflection, so a reflected reconstruction has exactly the same internal distance geometry as the non-reflected structure.

With exact distances, changing the reference atom would only change the coordinate frame, so the aligned reconstruction would still match the original structure essentially exactly. With noisy distances, however, there is an additional ambiguity: no exact 3D structure satisfies all noisy distances simultaneously. The SVD truncation to the three largest singular values therefore gives a best rank-3 approximation, and this approximation can depend slightly on the chosen reference atom.

Conclusion: using the same noisy distance matrix but changing the reference atom from atom 1 to atom 25 produced another full-rank noisy Gram matrix and a very similar c-RMSD. The c-RMSD changed from 1.447742892514 Å to 1.397538651934 Å, a difference of 0.050204240580 Å. This confirms that the main loss of exact reconstruction is due to the noisy distances, while the reference-atom choice only slightly affects the best rank-3 approximation.

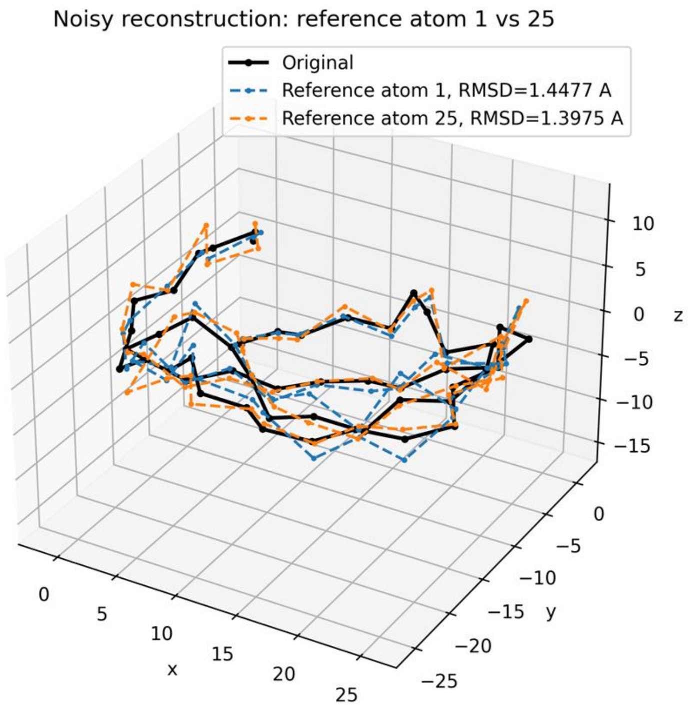

Figure 3. Comparison of noisy reconstructions using two different reference atoms.

(d) Reconstruction from Partial Contact-like Distances

Dataset and contact definition

The same protein fragment was used as in the previous parts: the Cα coordinates of residues 102–152 from chain A of PDB structure 6LU7. This gives n = 51 Cα atoms. Atom 1, corresponding to residue 102 Cα, was used as the reference atom. In this part, instead of using all pairwise distances, only a subset of contact-like distances was considered:

$$R = \{d_{ij}: \ |i - j| > 5, \ d_{ij} < 8 \text{ Å}\}$$

Thus, local backbone-neighbor distances were excluded from R by the condition $|i - j| > 5$, while only spatially close Cα pairs with distance below 8 Å were retained. For this 6LU7 fragment, the total number of distances in R was 72.

Sampling and distance completion

Two random subsets of R were used: 50% and 75%. A fixed random seed, 42, was used for reproducibility. The subsets were sampled using one fixed random permutation, so the 50% subset is contained within the 75% subset. This makes the comparison cleaner, because the 75% case contains the 50% contacts plus additional distance constraints.

Since the Gram matrix construction requires a complete pairwise distance matrix, the missing distances were approximated using shortest-path completion on a weighted graph. The retained contact distances were used as graph edges. Consecutive Cα–Cα backbone edges were also added as a connectivity scaffold, because the contact set R alone may be disconnected for this fragment.

$$\hat{d}_{ij} = \text{min path length from } i \text{ to } j$$

This gives an approximate completed distance matrix for each case. These completed distances are not the original exact all-pairs distances; they are approximations inferred from the retained partial distance information.

Gram matrix construction and coordinate reconstruction

For each completed distance matrix, an approximate Gram matrix was constructed using the same distance-geometry formula as before:

$$G_{ij} = \frac{\hat{d}_{0i}^2 + \hat{d}_{0j}^2 - \hat{d}_{ij}^2}{2}$$

where atom 0 is the reference atom, i.e. atom 1 in the assignment numbering. The resulting Gram matrix has shape 50 × 50 because the reference atom is placed at the origin and the remaining 50 Cα atoms are represented relative to it.

The coordinates were reconstructed by computing the SVD of the Gram matrix and retaining only the three largest singular values:

$$X_{\text{rec}} = U_3 \Sigma_3^{1/2}$$

This gives a three-dimensional rank-3 reconstruction. The reconstructed coordinates were then optimally aligned to the original Cα coordinates using Kabsch alignment, allowing reflection, and the coordinate RMSD was computed as:

$$c\text{−RMSD} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \|x_i - y_i\|^2}$$

Results

Subset	Contacts from R	Backbone scaffold edges	Numerical rank	det(R)	c-RMSD (Å)
50%	36	50	46	-1.0	10.8227

75%	54	50	46	-1.0	9.8968

For the 50% contact subset, 36 contact distances from R were retained. After adding the 50 consecutive backbone scaffold edges and completing the missing distances by shortest paths, the Gram matrix had numerical rank 46 using tolerance 10^{-8}. The reconstruction using the three largest singular values gave c-RMSD = 10.82270143926 Å after optimal alignment.

For the 75% contact subset, 54 contact distances from R were retained. The same 50 consecutive backbone scaffold edges were added, and the missing distances were again completed by shortest paths. The Gram matrix also had numerical rank 46 using tolerance 10^{-8}. The reconstruction using the three largest singular values gave c-RMSD = 9.896757056291 Å after optimal alignment.

Leading singular values

k	50% subset σ_k	75% subset σ_k
1	8.537067748806e+04	8.233676017478e+04
2	2.745741419591e+03	2.140773031802e+03
3	1.710956738149e+03	1.798896736642e+03
4	7.792170318641e+02	6.674779636096e+02
5	7.622045304516e+02	5.125350167374e+02
6	5.907571968634e+02	3.968842953847e+02
7	4.784410522366e+02	3.488292919912e+02
8	4.268427142539e+02	2.797428494624e+02
9	3.990552259242e+02	2.184407166493e+02
10	2.380499906008e+02	1.710487847370e+02

Comparison and interpretation

The 75% contact subset performed better than the 50% contact subset. The c-RMSD decreased from 10.82270143926 Å to 9.896757056291 Å, an improvement of 0.9259443829741 Å.

This behavior is expected because the 75% subset contains more contact constraints than the 50% subset. More retained distances give more geometric information, so the shortest-path completed distance matrix is generally closer to the true distance geometry. However, this improvement is not mathematically guaranteed for every possible random sample; it is the expected trend.

Both c-RMSD values are much larger than in the previous parts because here the reconstruction is based only on a partial set of contact-like distances, not on all pairwise distances. The shortest-path completion provides only approximate missing distances. Therefore, the reconstructed geometry is a rough approximation of the original Cα backbone, and the final c-RMSD remains relatively high.

The numerical rank was 46 in both cases, not 3. This is expected because the completed distance matrices obtained from partial contact information and shortest paths are not exact Euclidean distance matrices for points in R^3. The SVD truncation to the three largest singular values therefore gives a best rank-3 approximation rather than an exact reconstruction.

The determinant of the optimal alignment matrix was −1 in both cases, indicating that the best orthogonal alignment included a reflection. This is not an error: distance-based reconstruction is invariant under reflection, so the reconstructed structure may be mirror-equivalent to the original before alignment.

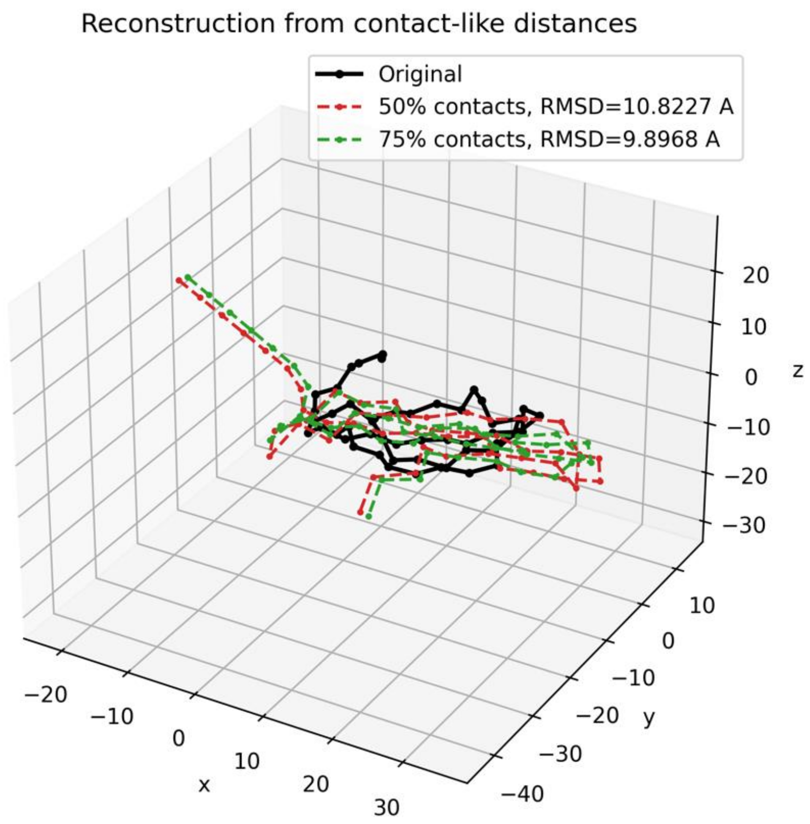

Figure 4. Reconstruction from partial contact-like distances. Original Cα backbone compared with reconstructions obtained from 50% and 75% randomly retained contact-like distances. The 75% subset gives a lower c-RMSD than the 50% subset, consistent with the use of more distance constraints.

<u>Conclusion</u>

Using only partial contact-like distances produces a substantially less accurate reconstruction than using complete distance information. The 50% subset gave c-RMSD = 10.82270143926 Å, while the 75% subset gave c-RMSD = 9.896757056291 Å. Thus, retaining more contact distances improved the reconstruction by approximately 0.9259443829741 Å. This confirms that the amount of retained distance information strongly affects the quality of the backbone reconstruction.

(e) Optimization refinement from partial contact reconstructions

Starting from the Gram/SVD reconstructions obtained in part (d), I refined the coordinates of the Cα backbone by solving the proposed optimization problem. The same PDB fragment was used: PDB structure 6LU7, chain A, residues 102–152, using only the Cα atoms. Therefore, the number of points was n = 51. The reference atom was atom 1, corresponding to the Cα atom of residue 102.

The contact set R was the same as in part (d). It contained contact-like long-range Cα distances satisfying |i − j| > 5 and d_ij < 8 Å. The total number of contacts in R was 72. For the two cases from part (d), I used the sampled 50% and 75% subsets of R as interval constraints, not all possible contacts.

<u>*Distance bounds and objective function*</u>

For every sampled contact distance d_ij in R, I defined lower and upper bounds as:

$$l_{ij} = d_{ij} - 0.2\ \text{Å}, \qquad u_{ij} = d_{ij} + 0.2\ \text{Å}$$

Thus, each sampled contact was allowed to vary within ±0.2 Å of its original value. The optimization problem minimized the sum of contact-bound penalties and a local backbone regularization term:

$$\min_X \sum_{d_{ij} \in R} \phi_{ij}(\|x_i - x_j\|_2) + \lambda \sum_{k=1}^{n-1} \psi_k(\|x_{k+1} - x_k\|_2), \qquad \lambda \geq 1$$

For a contact distance δ = ||x_i − x_j||_2, the contact penalty was:

$$\phi_{ij}(\delta) = \max(0, l_{ij} - \delta)^2 + \max(0, \delta - u_{ij})^2$$

This penalty is zero when δ lies inside the interval [l_ij, u_ij]. If δ is smaller than l_ij or larger than u_ij, the violation is penalized quadratically.

The local backbone regularization term was:

$$\psi_k(\delta) = (\delta - b_k)^2, \qquad b_k = \|x_{k+1}^{\text{orig}} - x_k^{\text{orig}}\|_2$$

Here, b_k is the true distance between consecutive Cα atoms k and k+1 in the original backbone. This term keeps the local backbone geometry realistic. I used λ = 10.0, which lies within the requested range 1–100.

Optimization setup

The initial coordinates for the optimization were the reconstructed coordinates from part (d), obtained from shortest-path distance completion followed by Gram matrix construction and SVD reconstruction. I used L-BFGS-B optimization. After refinement, the coordinates were optimally aligned to the original Cα coordinates before computing c-RMSD.

The coordinate RMSD was computed as:

$$c\text{−RMSD} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \|x_i - y_i\|^2}$$

For each sampled contact, I also computed the amount by which the refined distance violated its interval bounds:

$$v_{ij} = \max(0, l_{ij} - \delta_{ij}, \delta_{ij} - u_{ij})$$

The reported violation count is the number of sampled contact constraints with v_ij > 0.5 Å.

Numerical results

Case	**Sampled constraints**	**Initial c-RMSD (Å)**	**Refined c-RMSD (Å)**	**Violations > 0.5 Å**	**Optimization success**
50% contacts	36	10.82270143926	8.818679240482	0	True

75% contacts	54	9.896757056291	8.153294971332	0	True

For the 50% contact subset, the initial Gram/SVD reconstruction from part (d) had c-RMSD = 10.82270143926 Å. After optimization refinement, the c-RMSD decreased to 8.818679240482 Å. The optimization converged successfully after 79 iterations, with final objective value 1.639007447463 × 10^−12. The number of contact-bound violations larger than 0.5 Å was 0 out of 36 sampled constraints.

For the 75% contact subset, the initial Gram/SVD reconstruction from part (d) had c-RMSD = 9.896757056291 Å. After optimization refinement, the c-RMSD decreased to 8.153294971332 Å. The optimization converged successfully after 218 iterations, with final objective value 3.980948586484 × 10^−12. The number of contact-bound violations larger than 0.5 Å was 0 out of 54 sampled constraints.

Comparison of the two cases

The refinement improved both reconstructions. The 50% case improved from 10.8227 Å to 8.8187 Å, while the 75% case improved from 9.8968 Å to 8.1533 Å. The 75% refined reconstruction was better than the 50% refined reconstruction, as expected, because the 75% subset contained more sampled contact constraints. More distance constraints provide more geometric information and reduce the space of possible conformations.

However, even after refinement, the c-RMSD values remained relatively high compared with the exact-distance reconstruction. This is expected because the constraints are sparse: only a subset of long-range contact-like distances is used, not the full pairwise distance matrix. The optimization can satisfy the sampled interval constraints and the local backbone distances, but this does not guarantee recovery of the complete global fold.

Geometric ambiguity from sparse constraints

Sparse distance constraints may admit multiple reconstructions even when all sampled constraints are satisfied. The reason is that a small subset of pairwise distances does not uniquely determine all relative positions of the points in 3D space. Different global conformations can satisfy the same contact intervals, especially if the long-range contacts are few, unevenly distributed, or do not sufficiently constrain the global shape of the backbone.

In this optimization, the backbone term indirectly enforces local consecutive Cα–Cα geometry through the penalty on ||x_{k+1} − x_k||_2. This reduces unrealistic local distortions, but it still does not fully determine the global structure. Therefore, zero large bound violations do not necessarily imply low c-RMSD: a structure can satisfy all sampled contact intervals and local backbone distances while still differing globally from the original fold.

Conclusion

Using the output of part (d) as initialization, optimization refinement reduced the c-RMSD for both the 50% and 75% contact subsets. For 50% contacts, the c-RMSD decreased from 10.82270143926 Å to 8.818679240482 Å. For 75% contacts, it decreased from 9.896757056291 Å to 8.153294971332 Å. In both cases, the number of contact-bound violations larger than 0.5 Å was zero. The 75% case performed better, consistent with the fact that it contains more geometric constraints. Nevertheless, sparse contact constraints remain insufficient to uniquely determine the complete 3D backbone geometry.

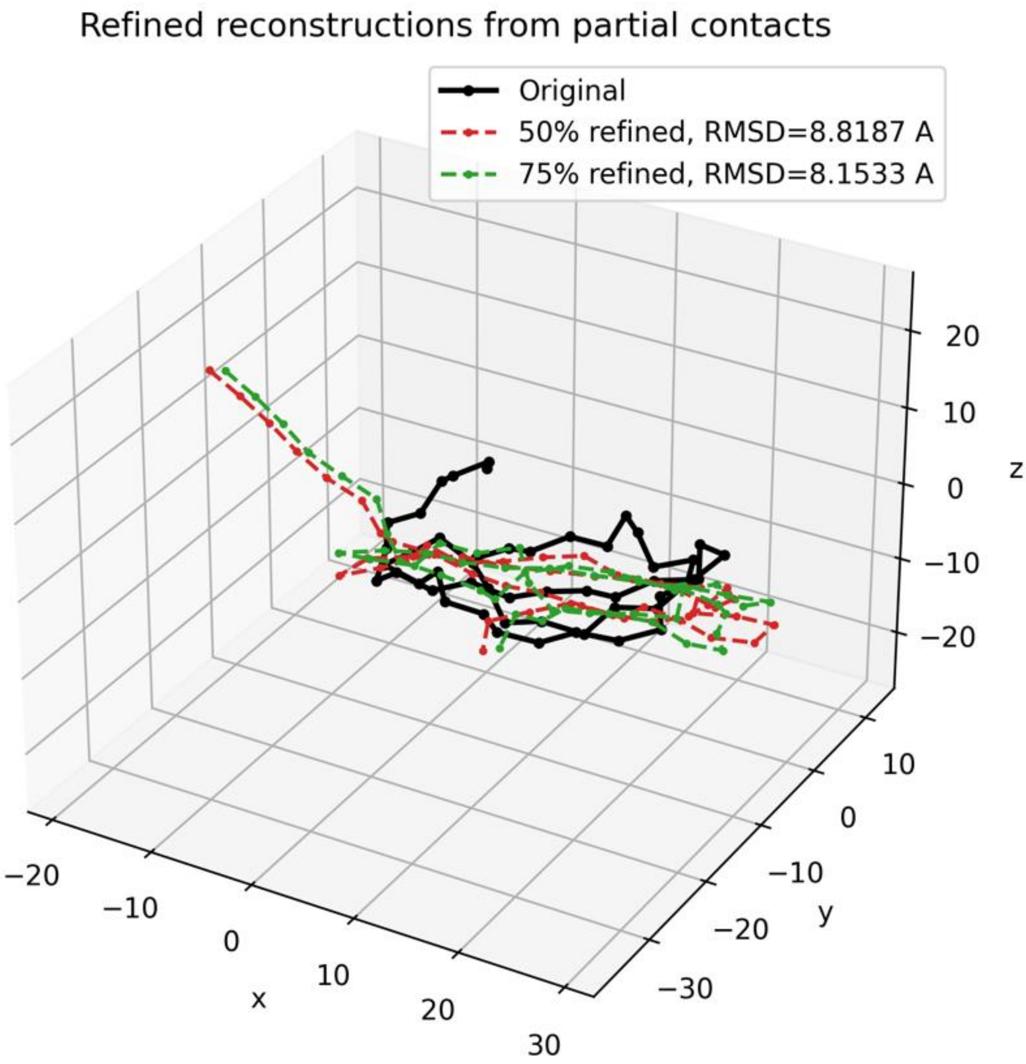

Figure 5. Refined reconstructions from partial contacts. The black curve shows the original Cα backbone. The red curve shows the refined reconstruction from the 50% contact subset, with c-RMSD = 8.8187 Å. The green curve shows the refined reconstruction from the 75% contact subset, with c-RMSD = 8.1533 Å.